MD ARSH AZAM

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PROFESSIONAL SUMMARY

Entry-level Python/Django Backend Developer with hands-on experience building database-driven web applications and RESTful APIs using Django and Django REST Framework. Experienced in authentication, role-based permissions, PostgreSQL, Django ORM, file uploads, email notifications, and automated testing. Built foundational Generative AI applications using Python, Hugging Face, OpenAI APIs, LangChain, embeddings, vector databases, and Retrieval-Augmented Generation. Interested in developing secure backend services and practical AI-powered applications.

TECHNICAL SKILLS

- **Languages:** Python
- **Backend Development:** Django, Django REST Framework (DRF), RESTful APIs, Authentication, Authorization, Role-Based Access Control (RBAC), Custom Permissions, Django ORM, CRUD Operations
- **Generative AI & LLMs:** Generative AI, Large Language Models (LLMs), Hugging Face Transformers, OpenAI API, Prompt Engineering, Transformer Architecture
- **LLM Application Development:** LangChain, Retrieval-Augmented Generation (RAG), AI Agents, Function Calling, Embeddings, Semantic Search
- **Vector Databases:** ChromaDB, Pinecone, Vector Search
- **Databases:** PostgreSQL, SQLite
- **Machine Learning & NLP:** Text Classification, Tokenization, Feature Extraction, Text Summarization, Model Fine-Tuning
- **Frontend:** HTML5, CSS3, Flexbox, CSS Grid
- **Testing:** Django Test Framework, Unit Testing, API Testing
- **Tools:** Git, GitHub, Visual Studio Code

PROJECTS

Job Portal | Django, Django REST Framework

GitHub: https://github.com/mdarshazam21/job-portal-django

- Built a recruitment platform with separate candidate and recruiter workflows for job posting, job search, filtering, pagination, resume uploads, and application-status tracking.
- Developed RESTful APIs using Django REST Framework serializers, custom permissions, and object-level access control.
- Implemented email notifications using Django's email framework, triggered automatically by application-status updates.
- Created and maintained 23 automated tests using Django's test framework, covering models, views, API behaviour, authentication, duplicate applications, resume uploads, and object-level permissions.
- **Technologies:** Python, Django, Django REST Framework, PostgreSQL, HTML, CSS

Library Management System | Django, Django REST Framework

GitHub: https://github.com/mdarshazam21/Library-Management-System-Django

- Built a library management application for book cataloging, member management, borrowing, returns, inventory tracking, and due-date-based late-fee calculation.
- Implemented staff-only permissions for borrowing and return operations, with public catalog search, filtering, and pagination.
- Developed DRF endpoints with serializers, custom actions, and permission classes for borrowing workflows through web and API interfaces.
- **Technologies:** Python, Django, Django REST Framework, SQLite, HTML, CSS

Document Q&A Chatbot | Python, LangChain, Hugging Face/OpenAI, ChromaDB

- Built a document question-answering application using Retrieval-Augmented Generation to retrieve relevant content and generate context-aware responses.
- Processed uploaded documents using document loaders, text chunking, embeddings, and vector similarity search.
- Integrated LangChain prompt templates, retrieval chains, conversation memory, and an LLM API.
- Implemented source-aware responses and tested retrieval quality using representative user questions.
- **Technologies:** Python, LangChain, OpenAI API or Hugging Face, ChromaDB, embeddings, RAG

EDUCATION

- **B.Tech in Electronics & Communication Engineering (AI/ML Minor)**
  Maharaja Agrasen Institute of Technology, New Delhi | 2022-2026 | CGPA: 7.1/10
- **Class XII (PCM)** | Panchsheel Public School | 2020 | 67.2%
- **Class X** | Panchsheel Public School | 2018 | 76%